



CS39006: NETWORKS LABORATORY

ASSIGNMENT 2: UDP SOCKETS

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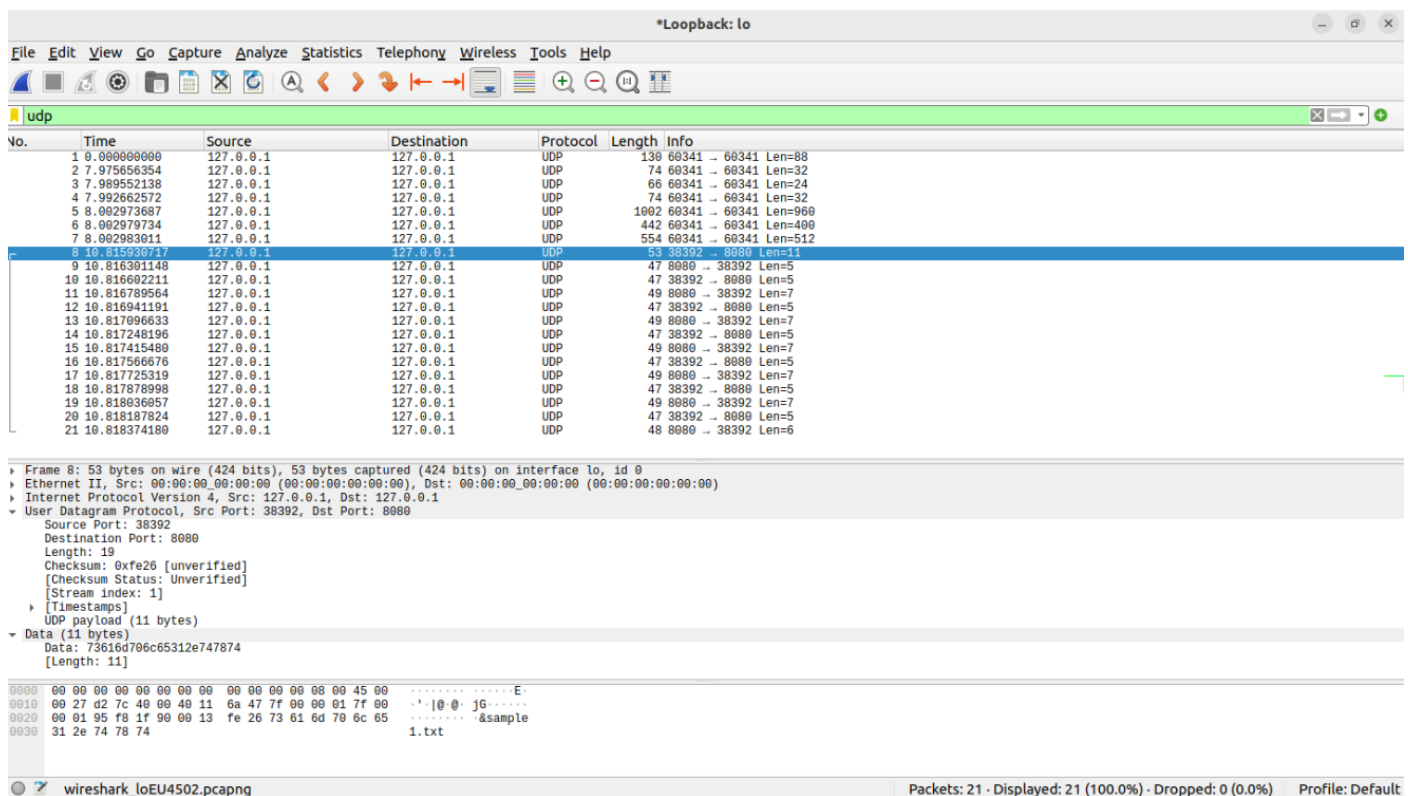
PART – 2: WIRESHARK ANALYSIS

1) Capture all packets exchanged between the client and server during execution. Show the screenshots.

Input File given:

1	HELLO
2	CS31206
3	CS39006
4	CS31208
5	CS11440
6	CS31207
7	FINISH

8	10.815930717	127.0.0.1	127.0.0.1	UDP	53	38392 → 8080	Len=11
9	10.816301148	127.0.0.1	127.0.0.1	UDP	47	8080 → 38392	Len=5
10	10.816602211	127.0.0.1	127.0.0.1	UDP	47	38392 → 8080	Len=5
11	10.816789564	127.0.0.1	127.0.0.1	UDP	49	8080 → 38392	Len=7
12	10.816941191	127.0.0.1	127.0.0.1	UDP	47	38392 → 8080	Len=5
13	10.817096633	127.0.0.1	127.0.0.1	UDP	49	8080 → 38392	Len=7
14	10.817248196	127.0.0.1	127.0.0.1	UDP	47	38392 → 8080	Len=5
15	10.817415480	127.0.0.1	127.0.0.1	UDP	49	8080 → 38392	Len=7
16	10.817566676	127.0.0.1	127.0.0.1	UDP	47	38392 → 8080	Len=5
17	10.817725319	127.0.0.1	127.0.0.1	UDP	49	8080 → 38392	Len=7
18	10.817878998	127.0.0.1	127.0.0.1	UDP	47	38392 → 8080	Len=5
19	10.818036057	127.0.0.1	127.0.0.1	UDP	49	8080 → 38392	Len=7
20	10.818187824	127.0.0.1	127.0.0.1	UDP	47	38392 → 8080	Len=5
21	10.818374180	127.0.0.1	127.0.0.1	UDP	48	8080 → 38392	Len=6



Packet No. **8 to 21 (both inclusive)** corresponds to the packets exchanged between the client and the server during the communication.

2) What protocol is used for communication?

The communication between the client and server is established using the **UDP (User Datagram Protocol)** in the transport layer. This is evident from the captured packets in Wireshark. The Protocol column in the Wireshark capture clearly shows UDP for all the packets exchanged between the client and server. The socket programming code uses **SOCK_DGRAM**, which is the socket type corresponding to UDP communication. This ensures that the messages sent by the client and server are encapsulated in UDP datagrams.

3) What are the source and destination IP addresses and ports?

From the Wireshark capture above, both the source and destination IP addresses are 127.0.0.1, which corresponds to the **localhost** i.e. loopback address. The loopback address is used for internal communication within the same machine, allowing a computer to send and receive data to itself. This is often used for testing purposes or local client-server communication. Regarding the ports:

Client Port: 38392 (shown as source for first packet, because it starts the communication)

Server Port: 8080 (shown as destination port, since it receives the filename)

4) What is the size (in bytes) of the FILENAME request sent by the client?

From the Wireshark capture, the packet containing the FILENAME request sent by the client has a UDP payload of **11 bytes**. This can be observed in the Packet Details section of the selected packet in Wireshark, which indicates the payload size.

```

> Frame 8: 53 bytes on wire (424 bits), 53 bytes captured (424 bits) on interface lo, id 0
> Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)
> Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
> User Datagram Protocol, Src Port: 38392, Dst Port: 8080
  Source Port: 38392
  Destination Port: 8080
  Length: 19
  Checksum: 0xfe26 [unverified]
  [Checksum Status: Unverified]
  [Stream index: 1]
  [Timestamps]
  UDP payload (11 bytes)
  Data (11 bytes)
    Data: 73616d706c65312e747874
    [Length: 11]

0000  00 00 00 00 00 00 00 00 00 00 00 00 08 00 45 00  .....E.
0010  00 27 d2 7c 40 00 40 11 6a 47 7f 00 00 01 7f 00  .'|@.@.jG
0020  00 01 95 f8 1f 90 00 13 fe 26 73 61 6d 70 6c 65  .....&sample
0030  31 2e 74 78 74                                     1.txt

```

The packet's total size (including Ethernet, IP, and UDP headers) is 53 bytes, as shown in the summary of the selected frame. However, the actual FILENAME "sample1.txt" request content (UDP payload) is only 11 bytes, which is the length of the data sent by the client (since it has 11 characters).

5) What is the size of the server's response for HELLO and the first word (WORD₁)?

The server's response containing HELLO is observed in a packet with a UDP payload size of **5 bytes**, as indicated in the Packet Details section. The server response is received when client sends the filename to the server. The packet's total size (including Ethernet, IP, and UDP headers) is 47 bytes, as shown in the summary of the selected frame.

```

> Frame 9: 47 bytes on wire (376 bits), 47 bytes captured (376 bits) on interface lo, id 0
> Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)
> Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
> User Datagram Protocol, Src Port: 8080, Dst Port: 38392
  Source Port: 8080
  Destination Port: 38392
  Length: 13
  Checksum: 0xfe20 [unverified]
  [Checksum Status: Unverified]
  [Stream index: 1]
  [Timestamps]
  UDP payload (5 bytes)
  Data (5 bytes)
    Data: 48454c4c4f
    [Length: 5]

0000  00 00 00 00 00 00 00 00 00 00 00 00 08 00 45 00  .....E.
0010  00 21 d2 7d 40 00 40 11 6a 4c 7f 00 00 01 7f 00  .!j@.@.jL
0020  00 01 1f 90 95 f8 00 0d fe 20 48 45 4c 4c 4f      .....HELLO

```

Once the client receives the response HELLO from server, it sends a packet containing **WORD1**. The client request containing **WORD1** is observed in a packet with a UDP payload size of **5 bytes**, as indicated in the Packet Details section. The packet's total size (including Ethernet, IP, and UDP headers) is 47 bytes, as shown in the summary of the selected frame.

```

> Frame 10: 47 bytes on wire (376 bits), 47 bytes captured (376 bits) on interface lo, id 0
> Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)
> Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
> User Datagram Protocol, Src Port: 38392, Dst Port: 8080
  Source Port: 38392
  Destination Port: 8080
  Length: 13
  Checksum: 0xfe20 [unverified]
  [Checksum Status: Unverified]
  [Stream index: 1]
  [Timestamps]
  UDP payload (5 bytes)
  Data (5 bytes)
    Data: 574f524431
    [Length: 5]

```

```

0000  00 00 00 00 00 00 00 00 00 00 00 00 08 00 45 00  .....E.
0010  00 21 d2 7e 40 00 40 11 6a 4b 7f 00 00 01 7f 00  ..!~@. jK.....
0020  00 01 95 f8 1f 90 00 0d fe 20 57 4f 52 44 31  .... WORD1

```

The server's response containing **CS31206** i.e. the first word of the input file, is observed in a packet with a UDP payload size of **7 bytes**, as indicated in the Packet Details section. The server response is received when client sends the request **WORD1** to the server. The packet's total size (including Ethernet, IP, and UDP headers) is 49 bytes, as shown in the summary of the selected frame.

```

> Frame 11: 49 bytes on wire (392 bits), 49 bytes captured (392 bits) on interface lo, id 0
> Ethernet II, Src: 00:00:00_00:00:00 (00:00:00:00:00:00), Dst: 00:00:00_00:00:00 (00:00:00:00:00:00)
> Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
> User Datagram Protocol, Src Port: 8080, Dst Port: 38392
  Source Port: 8080
  Destination Port: 38392
  Length: 15
  Checksum: 0xfe22 [unverified]
  [Checksum Status: Unverified]
  [Stream index: 1]
  [Timestamps]
  UDP payload (7 bytes)
  Data (7 bytes)
    Data: 43533331323036
    [Length: 7]

```

```

0000  00 00 00 00 00 00 00 00 00 00 00 00 08 00 45 00  .....E.
0010  00 23 d2 7f 40 00 40 11 6a 48 7f 00 00 01 7f 00  ..#..@. jH.....
0020  00 01 1f 90 95 f8 00 0f fe 22 43 53 33 31 32 30  .... "CS3120
0030  36                                     6

```

6) Inspect the payload of packets where the words are transmitted. Show the UDP payloads of those packets.

The server's response contains packets where the words are transmitted. A UDP payload size of **7 bytes**, as indicated in the Packet Details section, since it contains words of form **CSXXXXXX**. The packet's total size (including Ethernet, IP, and UDP headers) is 49 bytes, as shown in the summary of the selected frame. The exact payload details can be seen in the above image.

7) Measure the total time taken for the file transfer from start to finish

To calculate the total time taken for the file transfer, we look at the timestamps of the packets corresponding to the start and end of the communication between the client and server:

Start of Communication (Packet No. 8): The timestamp for Packet No. 8 is **10.815930717 seconds**.

End of Communication (Packet No. 21): The timestamp for Packet No. 21 **10.818374180 seconds**.

The total time taken for the communication can be calculated as:

Total Time = End Time – Start Time

Total Time = 10.818374180 – 10.815930717 = 0.002443463 seconds

Thus, the total time taken for the file transfer is **approximately 2.44 milliseconds (ms)**.

8) What is the average size of each packet during the communication?

From the provided Wireshark data, the sizes of the packets are:

8	10.815930717	127.0.0.1	127.0.0.1	UDP	53	38392 → 8080	Len=11
9	10.816301148	127.0.0.1	127.0.0.1	UDP	47	8080 → 38392	Len=5
10	10.816602211	127.0.0.1	127.0.0.1	UDP	47	38392 → 8080	Len=5
11	10.816789564	127.0.0.1	127.0.0.1	UDP	49	8080 → 38392	Len=7
12	10.816941191	127.0.0.1	127.0.0.1	UDP	47	38392 → 8080	Len=5
13	10.817096633	127.0.0.1	127.0.0.1	UDP	49	8080 → 38392	Len=7
14	10.817248196	127.0.0.1	127.0.0.1	UDP	47	38392 → 8080	Len=5
15	10.817415480	127.0.0.1	127.0.0.1	UDP	49	8080 → 38392	Len=7
16	10.817566676	127.0.0.1	127.0.0.1	UDP	47	38392 → 8080	Len=5
17	10.817725319	127.0.0.1	127.0.0.1	UDP	49	8080 → 38392	Len=7
18	10.817878998	127.0.0.1	127.0.0.1	UDP	47	38392 → 8080	Len=5
19	10.818036057	127.0.0.1	127.0.0.1	UDP	49	8080 → 38392	Len=7
20	10.818187824	127.0.0.1	127.0.0.1	UDP	47	38392 → 8080	Len=5
21	10.818374180	127.0.0.1	127.0.0.1	UDP	48	8080 → 38392	Len=6

Packet 8: 53 bytes (FILENAME)
 Packet 9: 47 bytes (HELLO)
 Packet 10: 47 bytes (WORD1)
 Packet 11: 49 bytes
 Packet 12: 47 bytes
 Packet 13: 49 bytes
 Packet 14: 47 bytes
 Packet 15: 49 bytes
 Packet 16: 47 bytes
 Packet 17: 49 bytes
 Packet 18: 47 bytes
 Packet 19: 49 bytes
 Packet 20: 47 bytes
 Packet 21: 48 bytes (FINISH)

Total Size = 53 + 47 x 7 + 49 x 5 + 48 = 675 bytes

Avg. Packet Size = 675 / 14 = 48.214285 bytes

Each Packet contains a Header of 42 bytes which includes Ethernet, IP, and UDP headers.

Therefore, the average payload of UDP Packets is **48.214285 – 42 = 6.214285 bytes.**

Packet Communication Summary:

- 1) Total number of packets: 14
- 2) Total size of packets: 675 bytes
- 3) Average size of each packet: **48.214285** bytes
- 4) Average payload of UDP Packet: **48.214285 – 42 = 6.214285 bytes.**